

Cyber Security Analyst Internship Program



Week 01 Assignment

Cybersecurity Fundamentals & Security Concepts

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Introduction

Cybersecurity has become one of the most important aspects of modern technology because individuals, businesses, and governments rely heavily on digital systems to store and exchange information. As cyber threats continue to increase in both frequency and sophistication, organizations must implement effective security measures to protect their data, systems, and networks.

This report explores the fundamental concepts of cybersecurity by examining real-world cyberattacks, analyzing how they affected the Confidentiality, Integrity, and Availability (CIA) Triad, and researching commonly used cybersecurity tools. It also includes the design of a secure home network and a discussion of the daily responsibilities of a cybersecurity analyst.

The objective of this assignment is to develop a strong understanding of cybersecurity fundamentals while improving research, documentation, and analytical skills. The knowledge gained from this assignment provides a solid foundation for future cybersecurity studies and practical experience in the field.

Research Five Real-World Cyberattacks

Attack Name	Year	Target Organization	Attack Type	Description	Impact
WannaCry Ransomware Attack	2017	United Kingdom National Health Service (NHS) and over 200,000 computers worldwide	Ransomware	WannaCry spread rapidly by exploiting a vulnerability in Microsoft Windows. It encrypted victims' files and demanded Bitcoin payments to restore access.	Hospitals cancelled appointments and surgeries, businesses suffered downtime, and financial losses exceeded billions of dollars globally.
Equifax Data Breach	2017	Equifax	Data Breach	Attackers exploited an unpatched web application vulnerability, allowing unauthorized access to sensitive customer information.	Personal information of approximately 147 million people was exposed, including Social Security numbers, birth dates, and addresses.
SolarWinds Supply Chain Attack	2020	SolarWinds and thousands of government agencies and private companies	Supply Chain Attack	Cybercriminals inserted malicious code into a software update distributed by SolarWinds, allowing attackers to infiltrate customers' networks.	Thousands of organizations, including government agencies, were compromised, leading to extensive espionage and security investigations.
Colonial Pipeline Attack	2021	Colonial Pipeline	Ransomware	The DarkSide ransomware group encrypted company systems, forcing Colonial Pipeline to temporarily shut down fuel distribution across the eastern United States.	Fuel shortages, panic buying, increased fuel prices, and millions of dollars in financial losses occurred.
Target Data Breach	2013	Target Corporation	Point-of-Sale (POS) Malware	Attackers gained access through a third-party vendor and installed malware on POS systems to steal	About 40 million payment card records and 70 million customer records were stolen, resulting in

Attack Name	Year	Target Organization	Attack Type	Description	Impact
				customers' payment card information.	financial losses and reputational damage.

These incidents demonstrate that cyberattacks can affect organizations of every size and industry. While some attacks focus on stealing confidential information, others disrupt critical services or compromise the software supply chain. They also highlight the importance of timely software updates, strong access controls, employee awareness, network monitoring, and effective incident response strategies.

CIA Triad Analysis

The **CIA Triad** is the foundation of information security. It consists of:

- **Confidentiality** – Protecting sensitive information from unauthorized access.
- **Integrity** – Ensuring data remains accurate, complete, and unaltered.
- **Availability** – Ensuring systems and data are accessible to authorized users whenever needed.

The table below explains how each cyberattack violated one or more components of the CIA Triad.

Cyberattack	CIA Component(s) Violated	Explanation
WannaCry Ransomware (2017)	Availability	WannaCry encrypted files and locked users out of their systems, preventing access to important data and services. Hospitals and businesses were unable to operate normally until systems were restored.
Equifax Data Breach (2017)	Confidentiality	Attackers gained unauthorized access to highly sensitive personal information, exposing the private data of approximately 147 million individuals.
SolarWinds Supply Chain Attack (2020)	Integrity and Confidentiality	Attackers modified legitimate software updates by inserting malicious code, compromising the integrity of the software. They also gained unauthorized access to confidential government and corporate information.

Cyberattack	CIA Component(s) Violated	Explanation
Colonial Pipeline Attack (2021)	Availability	The ransomware attack forced the company to shut down pipeline operations, making fuel distribution unavailable across much of the eastern United States.
Target Data Breach (2013)	Confidentiality	Attackers stole customers' payment card details and personal information, violating the confidentiality of sensitive financial data.

The five cyberattacks illustrate that different attacks can impact different aspects of the CIA Triad. Data breaches such as the **Equifax** and **Target** incidents primarily compromise **Confidentiality** by exposing sensitive information. Ransomware attacks like **WannaCry** and **Colonial Pipeline** mainly affect **Availability** because they prevent legitimate users from accessing systems and data. The **SolarWinds** attack demonstrates how attackers can compromise both **Integrity** by modifying trusted software updates and **Confidentiality** by using that access to steal sensitive information.

Understanding the CIA Triad helps cybersecurity professionals identify security risks, prioritize protective measures, and implement effective controls to reduce the impact of cyber threats.

Research Three Cybersecurity Tools

Cybersecurity professionals rely on specialized tools to identify vulnerabilities, monitor networks, detect threats, and respond to security incidents. The following are three widely used cybersecurity tools.

Tool Name	Purpose	Function	Real-World Use Case
Wireshark	Network traffic analysis	Captures and analyzes network packets in real time, allowing security professionals to inspect communication between devices and identify suspicious activity.	A security analyst uses Wireshark to investigate unusual network traffic and detect malware communication or unauthorized data transfers.
Nmap (Network Mapper)	Network discovery and security auditing	Scans networks to discover active hosts, open ports, running services, and operating systems. It also helps identify	During a security assessment, an analyst uses Nmap to identify open ports and services

Tool Name	Purpose	Function	Real-World Use Case
Metasploit Framework		potential security vulnerabilities.	that could be exploited by attackers.
	Penetration testing and vulnerability validation	Provides a collection of exploits, payloads, and auxiliary modules that help security professionals safely test systems for known vulnerabilities.	An ethical hacker uses Metasploit during an authorized penetration test to verify whether a discovered vulnerability can be exploited before recommending remediation.

Discussion

These tools play different but complementary roles in cybersecurity operations.

- **Wireshark** is mainly used for **network monitoring and traffic analysis**, helping analysts investigate suspicious network activity.
- **Nmap** is a **network reconnaissance** tool that identifies devices, services, and potential weaknesses within a network.
- **Metasploit Framework** is used for **penetration testing**, enabling security professionals to safely simulate cyberattacks and validate security vulnerabilities.

Together, these tools help organizations strengthen their security posture by detecting vulnerabilities, monitoring network activity, and testing defenses before malicious attackers can exploit weaknesses.

Task 4 – Secure Home Network Design

Secure Home Network Diagram

Internet

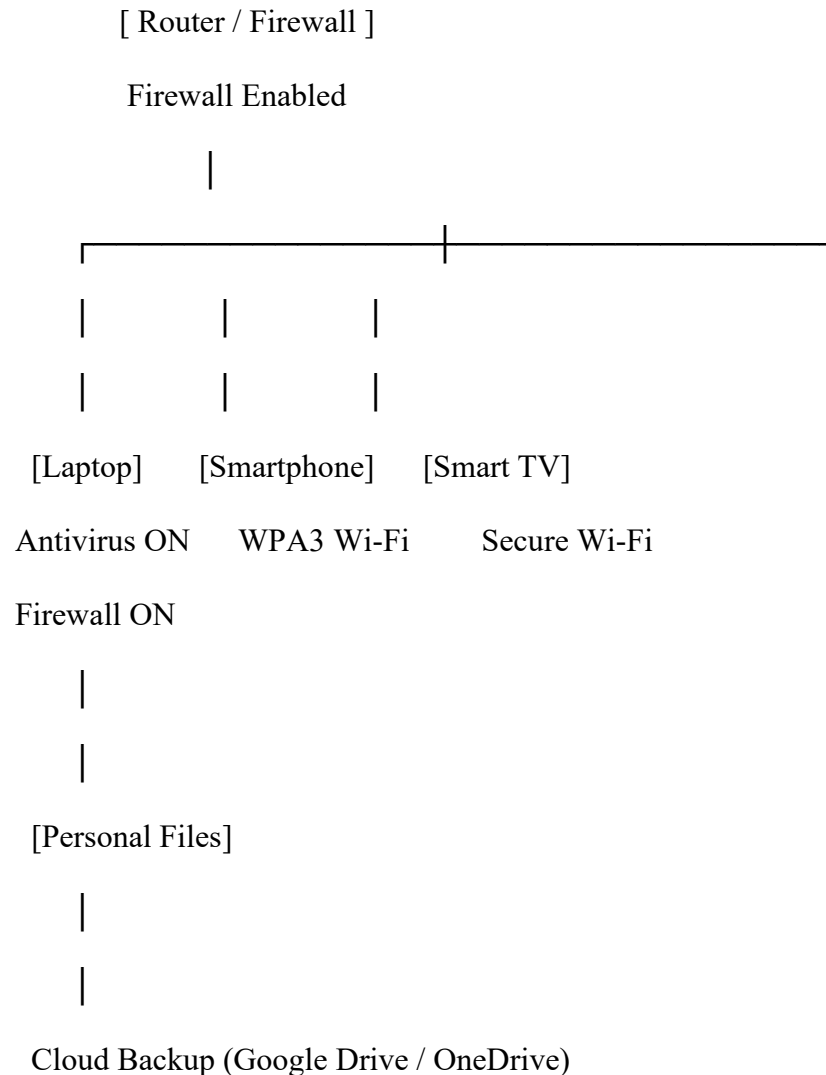
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[ISP Modem]

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Explanation of the Network Design

1. Internet

The Internet is the external network that provides connectivity to online services. Since it is a public network, it is the primary source of many cyber threats.

2. ISP Modem

The modem connects the home network to the Internet Service Provider (ISP). It converts signals between the ISP's network and the home network.

3. Router with Firewall

The router distributes internet access to all connected devices. Its built-in firewall filters incoming and outgoing traffic, blocking unauthorized access and helping protect the network from attacks.

4. Laptop/Desktop

The laptop is protected with:

- Antivirus software
- Operating system updates
- Local firewall
- Strong passwords
- Multi-factor authentication (where available)

5. Smartphone

The smartphone connects securely using **WPA3** or **WPA2** Wi-Fi encryption and should receive regular operating system and application updates.

6. Smart TV / IoT Devices

Internet-connected devices such as Smart TVs should use strong passwords, receive firmware updates, and be connected only to trusted networks to reduce security risks.

7. Cloud Backup

Important files should be backed up to a trusted cloud service or an external storage device. Regular backups help recover data after ransomware attacks, hardware failures, or accidental deletion.

8. VPN (Optional but Recommended)

A Virtual Private Network (VPN) encrypts internet traffic, protecting sensitive information when using public Wi-Fi or accessing online services remotely.

Security Best Practices

To maintain a secure home network, the following best practices should be followed:

- Use **WPA3** encryption (or **WPA2** if WPA3 is unavailable).
- Change the router's default administrator username and password.
- Enable the router's built-in firewall.
- Keep the router firmware updated.

- Install reputable antivirus software on all computers.
 - Enable automatic operating system updates.
 - Use strong, unique passwords for all accounts and devices.
 - Enable Multi-Factor Authentication (MFA) whenever possible.
 - Regularly back up important files.
 - Use a VPN when connecting through public Wi-Fi networks.
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Why This Design Is Secure

This home network design follows fundamental cybersecurity principles by layering multiple security controls. The firewall filters malicious traffic, antivirus software helps detect and remove malware, secure Wi-Fi encryption protects wireless communications, regular updates reduce known vulnerabilities, and cloud backups ensure important data can be recovered if a security incident occurs. The optional VPN adds another layer of protection by encrypting internet traffic.

Task 5 – What Does a Cybersecurity Analyst Do Every Day?

A Cybersecurity Analyst is responsible for protecting an organization's computer systems, networks, and sensitive information from cyber threats. Their primary objective is to identify security risks, prevent cyberattacks, and respond quickly to incidents to minimize damage. As cyber threats continue to evolve, cybersecurity analysts play a critical role in maintaining the confidentiality, integrity, and availability of organizational data and services.

One of the key daily responsibilities of a cybersecurity analyst is **monitoring network and system activity**. Analysts use Security Information and Event Management (SIEM) tools and other monitoring solutions to detect suspicious activities, unauthorized access attempts, malware infections, and unusual network traffic. They continuously review security alerts and determine whether an event is a genuine security incident or a false alarm.

Cybersecurity analysts also perform **vulnerability assessments** by scanning systems and networks for security weaknesses. Using tools such as Nmap and vulnerability scanners, they identify outdated software, misconfigurations, and exposed services that could be exploited by attackers. Once vulnerabilities are discovered, analysts work with IT teams to apply software updates, security patches, and configuration changes to reduce risk.

Another important responsibility is **incident response**. When a security incident occurs, analysts investigate the source of the attack, determine the extent of the damage, contain the threat, and help restore affected systems. They collect digital evidence, analyze logs, and document their findings to improve future security measures and support organizational learning.

Cybersecurity analysts are also responsible for **implementing and maintaining security controls**. They configure firewalls, antivirus software, endpoint protection systems, intrusion detection systems (IDS), and intrusion prevention systems (IPS). They regularly review security policies and ensure that systems comply with organizational security standards and industry best practices.

In addition to technical responsibilities, cybersecurity analysts help improve **security awareness** within an organization. They educate employees about phishing attacks, password security, social engineering, safe internet practices, and the importance of reporting suspicious activities. Since human error is a common cause of security breaches, employee awareness is an essential part of cybersecurity.

Documentation is another critical aspect of the role. Cybersecurity analysts prepare incident reports, maintain security documentation, record vulnerabilities, and recommend improvements to strengthen the organization's overall security posture. These reports help management understand security risks and support informed decision-making.

In conclusion, a cybersecurity analyst performs a wide range of responsibilities that include monitoring systems, identifying vulnerabilities, responding to incidents, implementing security controls, educating users, and maintaining security documentation. Their work helps organizations protect valuable information, maintain business continuity, and defend against constantly evolving cyber threats.

Remaining Sections

Now we only need these final parts:

Challenges Faced

During this assignment, several challenges were encountered. Researching real-world cyberattacks required reviewing multiple reliable sources to ensure the information was accurate and up to date. Understanding how each attack related to the CIA Triad required careful analysis. Designing a secure home network also required knowledge of networking concepts and security best practices. Despite these challenges, the assignment improved my understanding of cybersecurity fundamentals and strengthened my research and documentation skills.

Conclusion

This assignment provided a solid introduction to the fundamentals of cybersecurity. By researching major cyberattacks, analyzing the CIA Triad, exploring industry-standard cybersecurity tools, designing a secure home network, and understanding the daily responsibilities of a cybersecurity analyst, I gained valuable knowledge of the cybersecurity field. These concepts form the foundation for more advanced cybersecurity studies and practical experience. The skills developed during this assignment will support my growth as a cybersecurity professional and prepare me for future challenges in protecting digital systems and information.